

Solving the Rubik's Cube with A*

Artificial Intelligence and Reasoning, Lab 1

Mathias CHEBBAH Lamine BETRAOUI Bouna SALL

Universite Paris Dauphine, PSL
Master 1 MIAGE

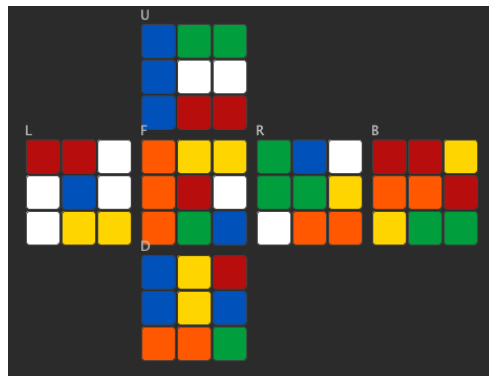
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The problem

- A Rubik's Cube has six faces of nine stickers.
- Each face can turn, which mixes the colors.
- The cube is solved when each face is one color.

Our goal: write the solve button.

Find a list of moves back to the solved cube.



The cube as a search problem

We use the five components of Lecture 2.

- **State:** a cube configuration.
- **Actions:** the twelve quarter turns.
- **Transition:** a turn permutes the stickers.
- **Goal test:** every face is uniform.
- **Cost:** each move costs one.

The shortest move list is the optimal solution.

The classes

We followed the statement closely, in Java.

- **Cube:** faces, the twelve turns, isSolved.
- **State:** a node, with g , h and expand.
- **Astar:** the solve method.
- **StateSet:** the explored set.
- **PriorityQueueState:** the frontier.
- **Heuristic:** the interface and its two versions.

The A* algorithm

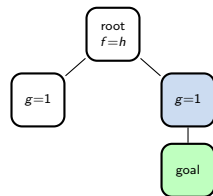
A* expands the node with the smallest value

$$f(n) = g(n) + h(n).$$

- $g(n)$ is the cost from the start.
- $h(n)$ estimates the cost to the goal.
- With an admissible h , A* is optimal.
- With $h = 0$, A* is a uniform cost search.

Uniform cost search and A*

- UCS orders the frontier by g .
- It is optimal but blind.
- A* adds the heuristic h .
- It looks toward the goal.
- So A* expands far fewer states.



Smallest f is expanded first.

Heuristic 1: the sticker relaxation

We assume each sticker moves on its own.

- Each sticker must reach its correct face.
- Distance is 0, 1 or 2 quarter turns.
- We sum over all 54 stickers.
- We divide by 12 stickers moved per turn.

$$h_{\text{label}} = \frac{1}{12} \sum_{s=1}^{54} d(s).$$

This is admissible, so A* stays optimal.

Heuristic 2: the piece relaxation

We assume each piece moves on its own.

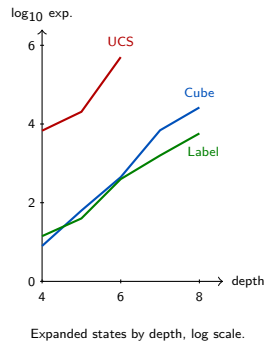
- A piece is a corner or an edge.
- One turn moves four corners and four edges.
- We count the misplaced pieces.

$$h_{\text{cube}} = \max \left(\left\lceil \frac{\text{misplaced corners}}{4} \right\rceil, \left\lceil \frac{\text{misplaced edges}}{4} \right\rceil \right).$$

This is also admissible.

Results

- UCS blows up around depth seven.
- Both heuristics scale much further.
- Plans stay optimal, same length.
- Memory is the first limit, not time.



- We solved the cube with A^* , as in the course.
- A^* returns an optimal solution.
- A good heuristic gives huge savings.
- UCS fails from about seven moves.
- The main limit of A^* is memory.

Thank you for your attention.